

Nathan Qiao

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EDUCATION

University of Western Ontario

BESc. in Mechanical Engineering & AI Systems Engineering (Dual Degree)

London, ON

Graduation Date: June 2028

Awards: Western Scholarship of Distinction (\$3500), ES1050 Best Implementation Award

University of California, Berkeley

Summer Exchange Student (Computer Science)

London, ON

June 2024 - August 2024

EXPERIENCE

Western University

August 2024 - Present

Residence Don - Engineering Living-Learning Community

London, ON

- Led a community of 35+ first year engineering students, **gathering feedback to identify common challenges** and implementing tailored academic and wellness initiatives that improved student engagement in events and academics.
- Developed a **centralized digital resource hub** to streamline access to residence and academic information, **securing adoption from Western Housing Leadership** and influencing residence-wide information sharing practice.
- Elected to the Residence Life Staff Advisory Committee (RLSAC) to represent over 35 Dons, **analyzing staff feedback and presenting policy recommendations** that enhanced training, communication, and resident support frameworks.

Siemens

May 2025 - August 2025

Business Analyst Intern

- Supported the **design and optimization of industrial valves and control systems** for fluid and steam applications, improving system performance, efficiency, and reliability across **3 major manufacturing clients**.
- Collaborated with **cross-functional teams** to prepare **10+ technical drawings and specification packages**, streamlining client integration and ensuring compliance with engineering and safety standards.
- Developed and presented **product performance reports** to leadership and client stakeholders that **articulated product value propositions** and total cost of ownership, **enhancing client buy-in** during procurement discussions.
- Supported on-site client demonstrations and **user feedback sessions**, capturing insights that shaped feature recommendations and long-term **customer engagement strategies**, leading to **2 successful system adoptions**.

PROJECTS

AI Driven Behavioral Analytics for Mental Wellbeing | Western University

September 2025 – December 2025

- Designed an end-to-end analytical framework in **Python** using **pandas** and **scikit learn** to evaluate whether music listening behavior contains a stable predictive signal for mental health severity.
- Led exploratory data analysis and **feature engineering to reduce noise**, address severe **class imbalance**, and evaluate whether nonlinear relationships could be learned from behavioral survey data.
- Experimented with **regression, classification, and tree-based models** including **XGBoost, CatBoost, and neural networks**, using **cross validation** and Optuna based **Bayesian optimization** to compare robustness and performance.
- Communicated **model limitations, statistical validity, ethical considerations**, and insights through clear visualizations and technical storytelling for non-technical audiences.

UFC Fantasy League Analytics Platform | Independent Data and Systems Project

August 2025 - Current

- Designed and implemented a **relational MySQL** database to store fighters, events, match outcomes, and user rosters.
- Built **Python data pipelines** to ingest, clean, and transform historical fight statistics from multiple **APIs** for analysis.
- Used **generative AI and AI/ML techniques to analyze** historical trends, evaluate feature options, and support **data-driven** drafting **strategies**, translating analytical findings into actionable strategies.
- Managed project codebase using **Git** for version control, collaboration, and reproducibility, ensuring traceable workflow.

LEADERSHIP EXPERIENCE

BTO London | Web Developer

September 2024 - Present

- Built and maintained the organization's website to improve user experience**, resulting in an increase in website visits, introducing more people to the organization's values and community support initiatives.
- Developed secure contact forms and event registration tools, **streamlining user interactions and reducing manual administrative workload** for staff.
- Enhanced site accessibility on mobile**, improving usability across devices and expanding reach to **underserved users**.

Western University | Foundations of Engineering (ES1050) CAD Student Mentor

January 2024 - Present

- Led weekly tutorial sessions and individualized office hours for **150+ first year engineering students**, simplifying complex CAD workflows and **resolving software issues** to improve student proficiency and confidence.
- Partnered with course instructors to redesign teaching materials, improving clarity and relevance, which **reduced assignment resubmission rates by 20%** and boosted **student engagement and performance**.

SKILLS & INTERESTS

Skills: Python, Java, Go, SQL, Cybersecurity Fundamentals, Data Analytics, MS Office, API Integration, Web Development

Interests: Camping, Portaging, MMA/Jiu-jitsu, Swimming, Rock Climbing, Chess, Mentorship, Traveling, Cooking, Music